

**Design and optimization of a stiff-hulled boat
powered by periodic internal motion**

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Registration number: Phy-120

Name of team member: Chen Yanlin

School: NUS High School of Math and Science

City, Country: Singapore, Singapore

Name of supervising teacher: Mr Tan Guan Seng

Position: Physics Teacher

School: NUS High School of Math and Science

City, Country: Singapore, Singapore

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Abstract

To date, various types of boats have been developed, yet there remains a lack of focus on boats which only interacts with the environment through its stiff hull. In such a boat, motion is constrained to within the hull, and the boat can only be powered through periodical mechanical movements of its internal parts. Here, a stiff-hulled boat is constructed, with periodic motion conferred by a linear actuator. A net force is achieved with an asymmetric shape, and an asymmetric velocity in the positive and negative directions. Under the assumption that the boat motion follows a quasiperiodic function at “steady state”, an analytic expression was derived for greater physical insight. The transient motion and average velocity under varying parameters were predicted both numerically and analytically, and extensively validated with experimental results. Optimization of various dimensionless parameters was performed, and a most optimized boat was predicted given constraints. It is intuitively expected that surface waves may affect the boat motion however, effects of such wave formation were shown to be negligible.

Keywords: Stiff-hulled boat; Quasiperiodic; Fluid dynamics; Airy wave theory

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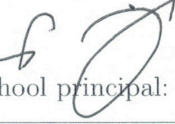
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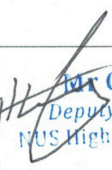
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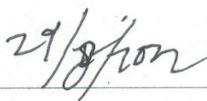
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1 Introduction

1.1 Background

Conventional boats are typically propelled by the turning of propeller blades, which leads to non-rigid hulls. For a boat that only interacts with the water through its stiff hull, any movements must be constrained to within the hull. As such, motion of the boat can only be conferred through asymmetry in the mechanical movements of the boat's internal parts, which will be termed as "internal movements" in this report.

Such stiff-hulled boats demonstrate various benefits: as they are enclosed and lacking in external propellers, the extent of wake and turbulence created is greatly reduced. Hence, these boats are optimal in situations where it is crucial to preserve the aquatic environment, such as in the case of on-going environmental studies.

1.2 Literature Review

Jones (1995) discussed the various types of boats that have been developed to date, which can largely be categorized under human-powered (and unpowered) boats, sailboats, or motorboats. Aside from unpowered crafts, such boats have a flexible hull in the form of oars, sails, or turbines.

Smith (2016) focused on propeller boats, a conventional type of motorboats. In such boats, The turning of propeller blades creates regions of high and low pressure. As water flow to the region of lower pressure, away from the boat, by conservation of momentum, the boat moves forward. Sun (2019) demonstrated that sailboats utilizes a similar propulsion method, where pressure differences and a transfer of momentum from air molecules to the boat produces the forward thrust force.

Sagerman (2020) investigated the effects of motorboat traffic on aquatic vegetation, and found a 58% reduction in vegetative abundance in areas with boat traffic. This was attributed to the effects of boating disturbance on aquatic beds, especially in shallower waters. The propellers of motorboats can directly cut or uproot vegetation, while the

wake and turbulence generated can disturb soft sediment bottoms. This stirring up of sediments will reduce the light intensity reaching benthic vegetation due to increased water turbidity, and poses a risk of smothering when sediment settle on the shoots.

In addition to these physical effects of propellers and increased water movement, Burgin (2011) stated that recreational motorboats pose chemical threats, through the release of chemical pollutants (such as fuel and lubricants) from combustion engines. These showcases the unique benefits of boats with a stiff, enclosed hull, thus highlighting the practical significance of this research.

1.3 Boat Design

As concluded previously, an asymmetry is necessary for net forward motion of the boat: to create the net external force, the drag force as the boat moves backwards must be greater than the drag experienced as the boat moves forward. To achieve this asymmetric drag, either the drag, which affects the drag coefficient, or the velocity at which the boat moves, can be modified.

This asymmetry can be realised even with a *periodic* internal motion (i.e. periodic motion of internal parts). A simple Direct Current motor is used, while a linear actuator converts the rotation of the motor into linear motion of the load. A large square tank (1.5 meters in length, 0.6 meters in depth) is used to stimulate a large water surface, such as in the case of lakes or oceans.

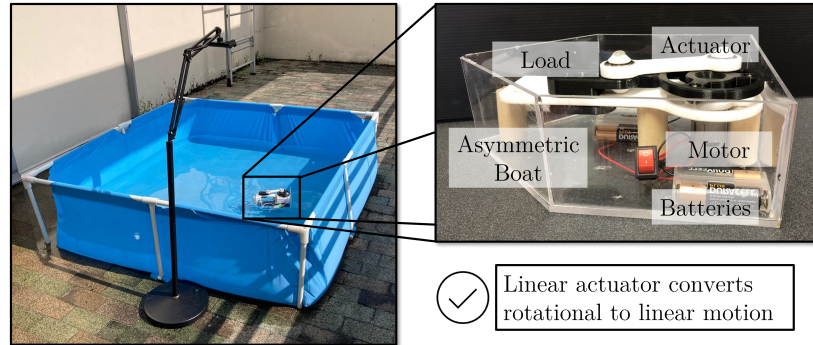


Figure 1: Experimental setup: large tank and boat with linear actuator

In Figure 1, the asymmetry in shape confers a difference in the forward and backward drag coefficient, while the horizontal actuator ensures a sinusoidal velocity of the load. To vary the angular frequency of the boat, the actuator is rotated to be vertical (Figure 2), such that the angular frequency as the load moves forward (and boat moves backwards) is higher than as the load moves backwards.

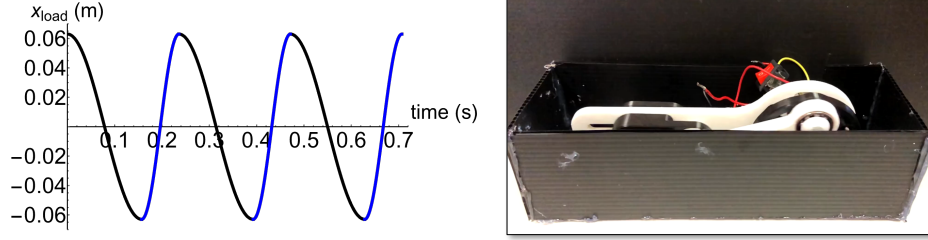


Figure 2: Experimental setup: symmetric boat shape with asymmetric angular frequency

1.4 Preliminary Observations

Aligning with intuitive expectations, a completely symmetrical boat remains stationary (Figure 3), while a boat with either asymmetric shape (Figure 4) or asymmetric angular frequency (Figure 5) displays net forward motion. The variable x_{boat} represents the position of the boat with respect to a stationary point in the lab frame.

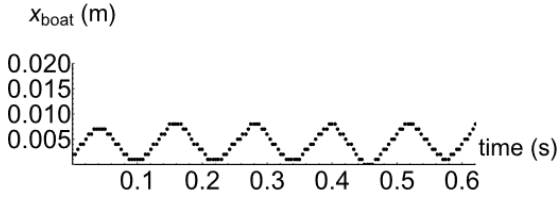


Figure 3: x - t graph of symmetric boat

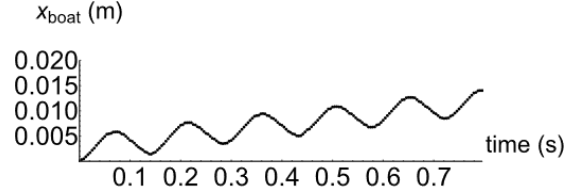


Figure 4: x - t graph of boat with asymmetric shape

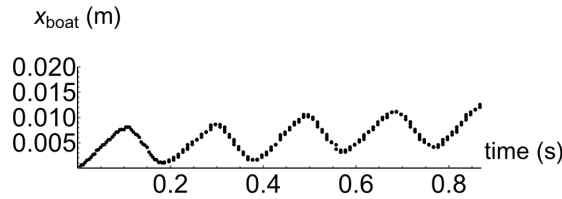


Figure 5: x - t graph of boat with asymmetric angular frequency

2 Quantitative Account

2.1 Numerical Simulation

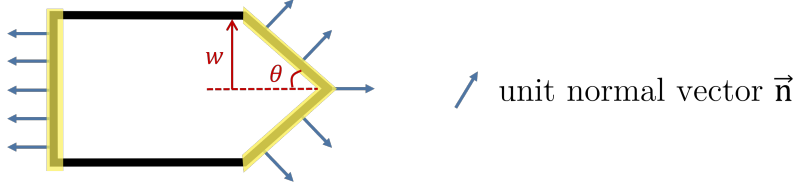


Figure 6: Schematic top-view of boat

Under turbulent conditions, only the component of velocity that is normal to the surface can create the drag, and this normal velocity $\vec{u}$ is defined as,

$$\vec{u} = \begin{cases} \dot{x}_{\text{boat}} \sin \theta \hat{n} & \text{for } \dot{x}_{\text{load}} < 0 \\ \dot{x}_{\text{boat}} \hat{n} & \text{for } \dot{x}_{\text{load}} \geq 0 \end{cases} \quad (2.1)$$

As the boat moves forward, the velocity that contributes to the drag force is integrated over the front of the boat, and similarly as the boat moves backwards. The drag force can be represented as the piecewise function

$$\vec{F}_{\text{drag}} = -\frac{1}{2} \rho \cdot c_D \int_{-w}^w d|\vec{u}|^2 dz \cdot \text{sgn}(\dot{x}_{\text{boat}}) \quad (2.2)$$

where ρ is the density of water, c_D is the drag constant, and d is the mean water depth. Since drag is the only external force, Newton's Second Law can be written as,

$$(m + M)\ddot{x}_{\text{COM}} = \vec{F}_{\text{drag}} \quad (2.3)$$

where m is the mass of the load, M is the mass of the boat, and

$$x_{\text{COM}} = x_{\text{boat}} + \frac{m}{m + M} x_{\text{load}} \quad (2.4)$$

The position of the load with respect to the boat is,

$$x_{\text{load}} = R \cdot \cos(\omega(t) \cdot t) \quad (2.5)$$

where R is the amplitude of motion and ω is the angular frequency.

2.2 Analytic Modelling

An analytic model will now be discussed for greater physical insight. The most significant parameter for a boat is its average velocity v_o , which is expected to be a constant at steady state. The drag force $\vec{F}_{\text{drag}}$ is two orders of magnitude below the driving force $|m\ddot{x}_{\text{load}}|$, hence assumption is made that

$$\ddot{x}_{\text{boat}} \approx -\frac{m}{m+M}\ddot{x}_{\text{load}} \quad (2.6)$$

and that

$$\dot{x}_{\text{boat}} \approx v_o - \frac{m}{m+M}\dot{x}_{\text{load}} \quad (2.7)$$

For a boat with an asymmetric angular velocity, the position of the mass is a piecewise function defined as

$$x_{\text{load}} = \begin{cases} R \cdot \cos(\omega_1 t) & \text{for } 0 \leq t \bmod T < \frac{\pi}{\omega_1} \\ R \cdot \cos(\omega_2(t - T)) & \text{for } \frac{\pi}{\omega_1} \leq t \bmod T < T \end{cases} \quad (2.8)$$

where T is the period and ω_1, ω_2 are the angular velocities of the load as it moves backwards and forwards, respectively.

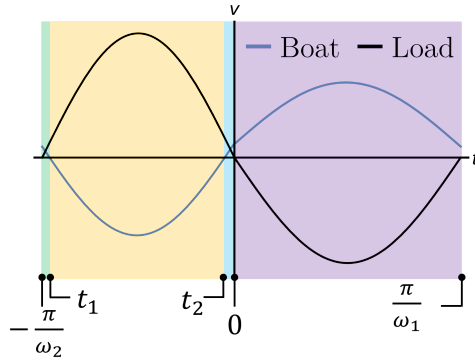


Figure 7: Velocity of boat (blue) and load (black)

For clarity, let $\gamma = \frac{mR}{m+M}$. Considering a single period spanning $-\frac{\pi}{\omega_2} \leq t \leq \frac{\pi}{\omega_1}$,

$$\dot{x}_{\text{boat}} = \begin{cases} \dot{x}_F = v_o + \gamma\omega_2 \sin(\omega_2 t) & \text{for } -\frac{\pi}{\omega_2} \leq t < 0 \\ \dot{x}_B = v_o + \gamma\omega_1 \sin(\omega_1 t) & \text{for } 0 \leq t < \frac{\pi}{\omega_1} \end{cases} \quad (2.9)$$

At steady state, the total force over the period is zero,

$$\begin{aligned} \int_{-\frac{\pi}{\omega_2}}^{\frac{\pi}{\omega_1}} F_D dt &\approx - \int_{-\frac{\pi}{\omega_2}}^{t_1} k_{\rightarrow} \dot{x}_F^2 dt + \int_{t_1}^{t_2} k_{\leftarrow} \dot{x}_F^2 dt \\ &- \int_{t_2}^0 k_{\rightarrow} \dot{x}_F^2 dt - \int_0^{\frac{\pi}{\omega_1}} k_{\rightarrow} \dot{x}_F^2 dt = 0 \end{aligned} \quad (2.10)$$

where $k_{\rightarrow}$ and $k_{\leftarrow}$ are the forward and backward drag coefficients such that

$$\vec{F}_{\text{drag}} = \begin{cases} k_{\rightarrow} \dot{x}_{\text{boat}}^2 & \text{for } \dot{x}_{\text{boat}} \geq 0 \\ k_{\leftarrow} \dot{x}_{\text{boat}}^2 & \text{for } \dot{x}_{\text{boat}} < 0 \end{cases} \quad (2.11)$$

Solving Equation 2.10 returns the expression for the average velocity¹,

$$v_o = \frac{2\gamma\omega_1\omega_2}{\pi} \frac{k_{\leftarrow} + k_{\rightarrow}}{k_{\leftarrow}\omega_1 - k_{\rightarrow}\omega_2} \left(1 - \sqrt{1 - \frac{\pi^2}{8\omega_1\omega_2} \frac{(k_{\leftarrow}\omega_2 - k_{\rightarrow}\omega_1)(k_{\leftarrow}\omega_1 - k_{\rightarrow}\omega_2)}{(k_{\leftarrow} + k_{\rightarrow})^2}} \right) \quad (2.12)$$

3 Experimental Verification

3.1 Phase Plots

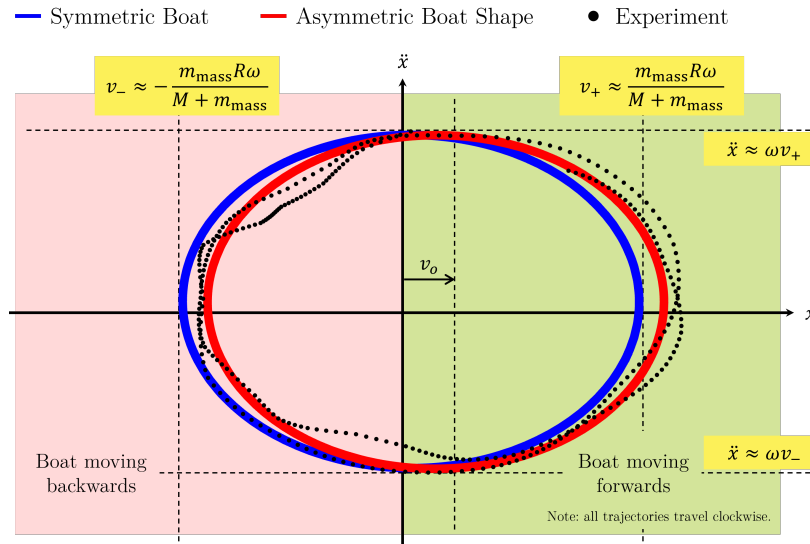


Figure 8: Phase plot for boat with asymmetric shape

¹The complete working is given in the Appendix.

To better understand the dynamics of the system, parametric phase plots were predicted theoretically and verified experimentally². Referring to Figure 8, at steady state, a completely symmetric boat follows oscillatory motion, and the acceleration-velocity graph resembles an ellipse. However, for a boat with an asymmetric shape, the drag when the boat is moving backward is higher than when the boat is moving forward. As such, the velocity in the forward direction is higher, and the ellipse is shifted to the right. The extent of the shift is given by v_o .

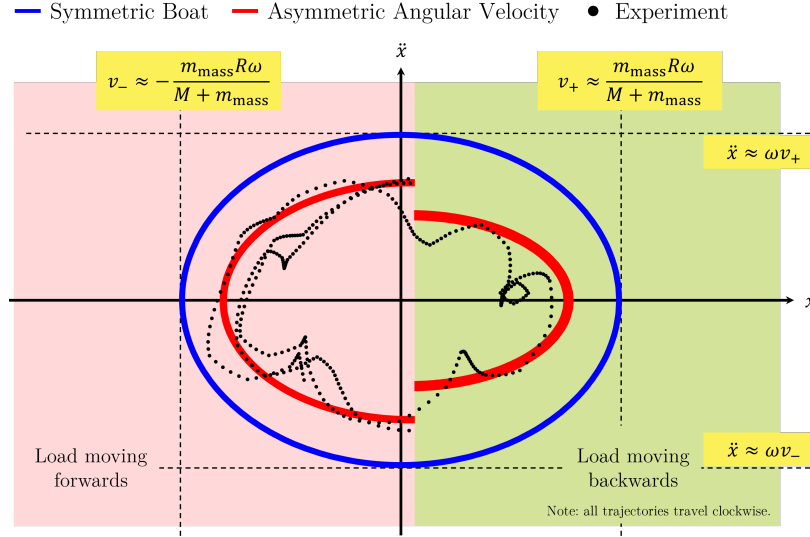


Figure 9: Phase plot for boat with asymmetric angular frequency

In contrast, for a boat with an asymmetric angular frequency (Figure 9), the angular frequency changes when the direction of load motion changes. Note that the velocity in the forward direction is less than the velocity in the backward direction, such that the boat experiences less drag when moving forward. Hence, the acceleration-velocity plot resembles two half-ellipses, where the positive half-ellipse has a smaller major and minor axis. Moreover, since the boat is moving forward with the average velocity, the velocity of the boat is only zero when the load is moving forward.

²To account for the noise that arises from differentiating discrete lists, the data is smoothed with moving average, interpolated, differentiated, and smoothed again.

3.2 Asymmetric Boat Shapes

The drag constant for the acrylic boat is first characterised from the decelerating curve in water, by fitting the experimental position-time graph to the function

$$m\ddot{x} = \frac{1}{2}\rho c_D A \dot{x}^2 \implies x(t) = k_2 - \frac{2m}{\rho c_D A} \ln \rho c_D A t + 2mk_1 \quad (3.1)$$

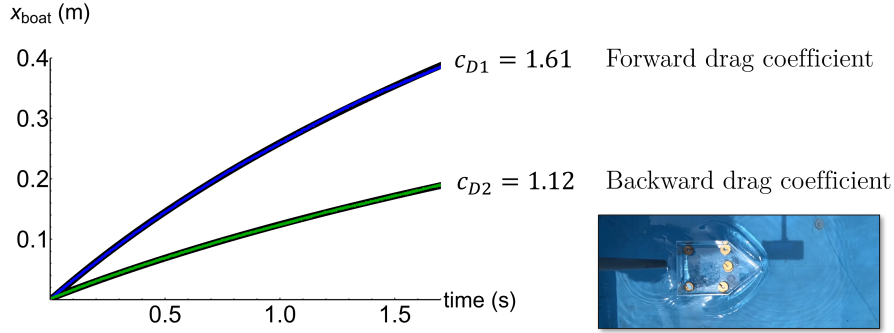


Figure 10: Characterisation for asymmetric acrylic boat

First, the effect of varying the shape of the boat on the net distance travelled is investigated, using two boats with different θ (Figure 11). Both transient trajectories were predicted theoretically and verified with experimental results. As expected, a smaller net distance travelled is seen for the boat with a less sharp tip, due to the smaller difference in drag coefficients in the forward and reverse direction.

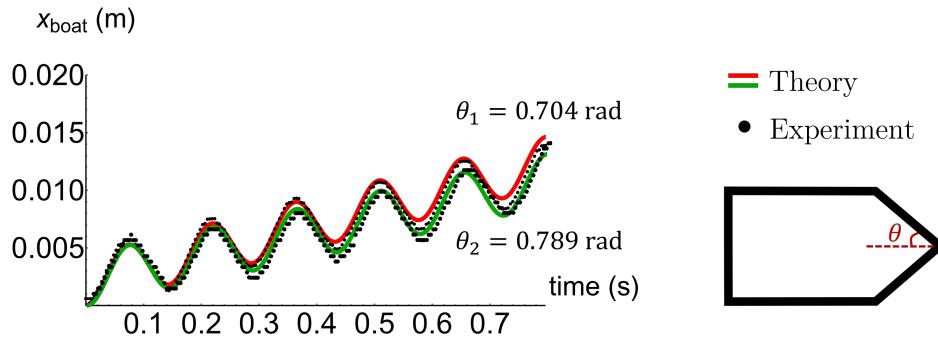


Figure 11: Transient graph for boat with asymmetric shape

To verify the accuracy of the analytic model, the average velocity is compared with both the numerical and analytic models.

	v_o , experimental	v_o , numerical	v_o , analytical
θ_1	0.0118 m/s	0.0115 m/s	0.0118 m/s
θ_2	0.0095 m/s	0.0081 m/s	0.0081 m/s

The mass of the boat is varied by adding load of known mass. As the mass increases, the amplitude of each oscillation decrease, leading to a lower velocity, which reduces the difference in drag. Hence, the net distance travelled and average velocity decreases.

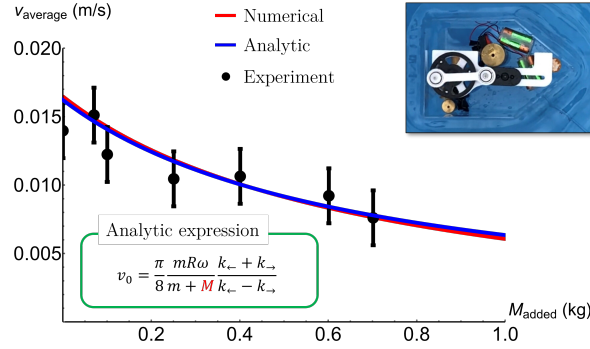


Figure 12: Experimental and theoretical overlay for different boat mass

As the angular frequency of the oscillations increase, the velocity of the boat in both forward and reverse directions increase, thus amplifying effects of the different drag coefficients and leading to a higher difference in drag force. This results in a higher average velocity.

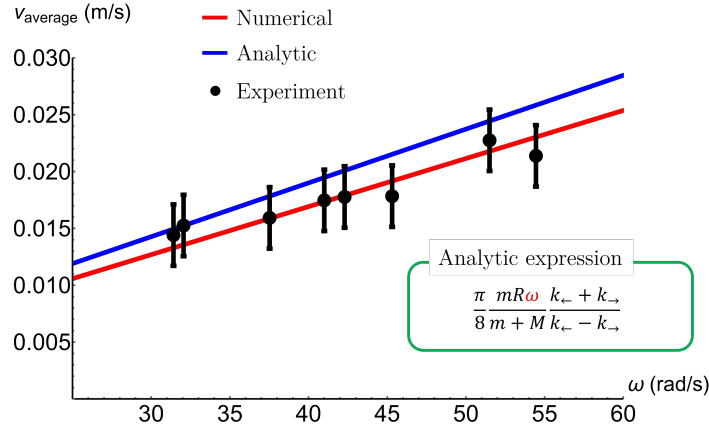


Figure 13: Experimental and theoretical overlay for different angular frequencies

3.3 Asymmetric Angular Frequencies

The drag constant for the boat with a symmetric shape is similarly characterised from the decelerating curve in water.

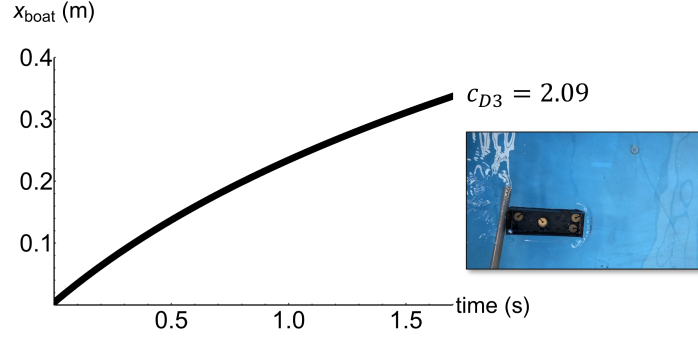


Figure 14: Characterisation for boat with symmetric shape

To verify the predictive power of the numerical model for a boat with asymmetric angular frequency, the transient motion is solved numerically and overlayed onto the experimental graph.

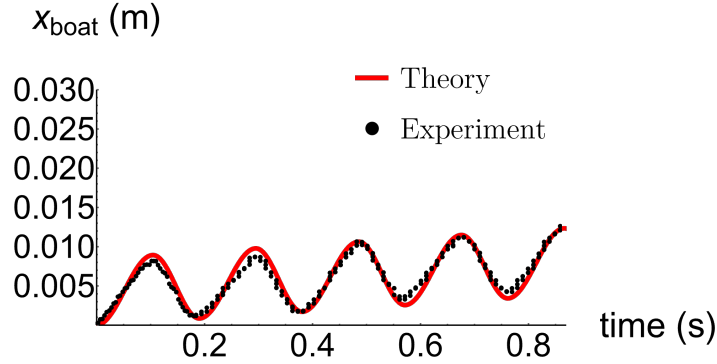


Figure 15: Transient graph for boat with asymmetric angular frequency

For a boat with a symmetric shape, when the angular frequency of the load moving forward (ω_2) is fixed, as ω_1 increases, the velocity of the boat moving forward increases. Hence, the hull experiences more drag when moving forward, and the average velocity decreases. Furthermore, since there is no difference in drag from boat shape, when the two angular frequencies are equal, at ratio equals one, the boat does not move forward, which agrees with what we had observed previously.

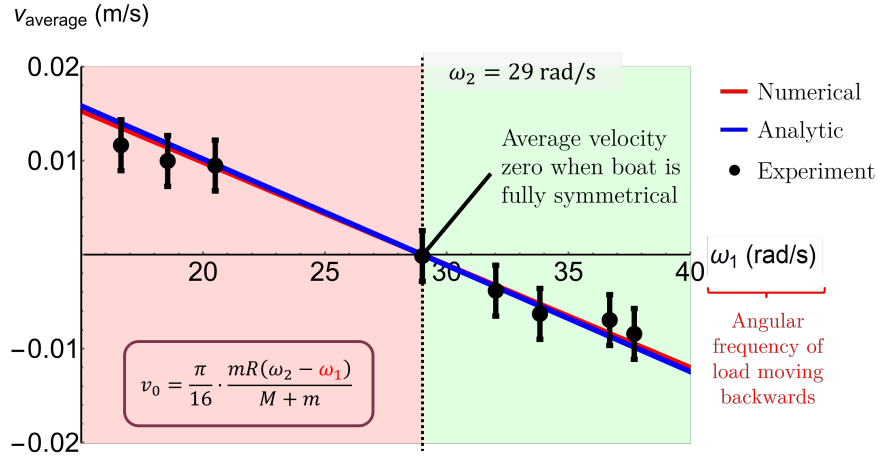


Figure 16: Experimental and theoretical overlay for varying angular frequency ratios with symmetric boat shape

The boat speed can be optimized by combining both an asymmetric boat shape and load speed. From Figure 17, a turning point is observed, since at high values of ω_1 , the boat experiences high drag moving forward, which reduced the average velocity, and at low values of ω_1 , each period will take a longer time to complete, which also reduces the average velocity.

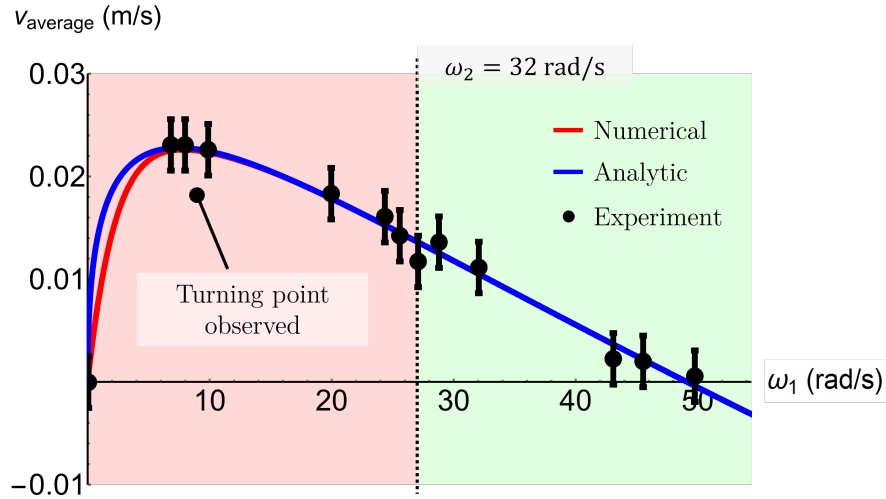


Figure 17: Experimental and theoretical overlay for varying angular frequency ratios with asymmetric boat shape

4 Optimization of Boat

To optimize the average velocity of the boat, the previous expression for the average velocity (Equation 2.12) can be re-expressed to

$$v_o = \frac{2\mu R\Omega_1\Omega_2}{\pi} \frac{k+1}{k\Omega_1 - \Omega_2} \left(1 - \sqrt{1 - \frac{\pi^2}{8\Omega_1\Omega_2} \frac{(k\Omega_2 - \Omega_1)(k\Omega_1 - \Omega_2)}{(k+1)^2}} \right) \quad (4.1)$$

which is in terms of 4 dimensionless parameters:

① Modified mass of load $\mu = \frac{m}{m+M}$	③ Modified backward ω_1 (of load) $\Omega_1 = \frac{\omega_1}{\omega_1 + \omega_2}$
② Ratio of drag coefficients $k = \frac{k_{\leftarrow}}{k_{\rightarrow}}$	④ Modified forward ω_2 (of load) $\Omega_2 = \frac{\omega_2}{\omega_1 + \omega_2}$

The average velocity increase monotonously with increasing modified mass of load, and ratio of drag coefficient, yet a turning point exists for varying values of forward and backward angular frequency. In particular, a relative maximum point is achieved when the backward angular frequency of the load is of a specific value; at that point, the most optimized boat is constructed for some given constraints (boat shape, mass, and total angular frequency).

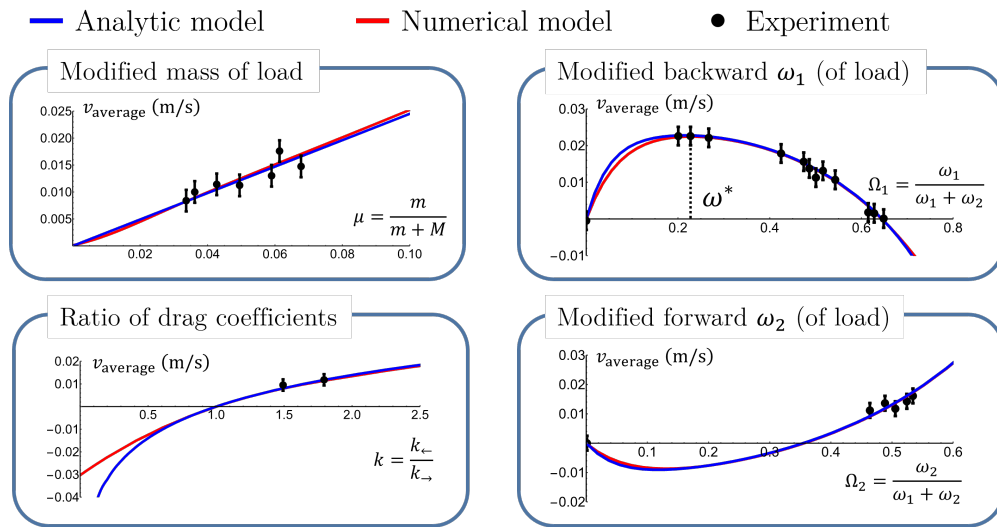


Figure 18: Experimental and theoretical overlay for varying dimensionless parameters

5 Effects of Gravity Wave Formation

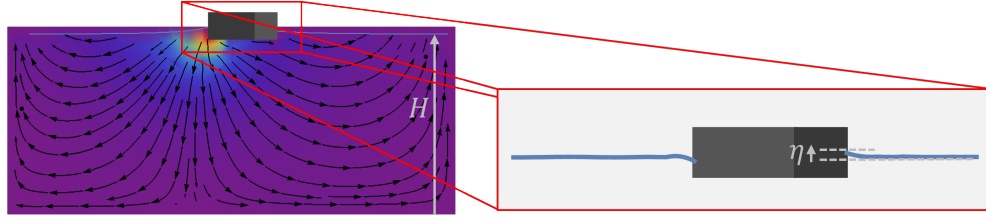


Figure 19: Airy wave theory

As the boat travels in water, gravity waves will form on the water surface. Such waves may change the depth at which the boat is immersed in water. To account for this, the airy wave theory (Landau, 1987) is used to predict the formation of gravity waves. To find the velocity potential ϕ , the Laplace equation

$$\nabla^2 \phi = 0 \quad (5.1)$$

is solved with the boundary conditions

$$\frac{\partial^2 \phi}{\partial t^2} + g \frac{\partial \phi}{\partial y} = 0, \text{ for } y = H \quad (5.2)$$

$$\frac{\partial \phi}{\partial y} = 0, \text{ for } y = 0 \quad (5.3)$$

Thereafter, the height of the wave, η , is found with

$$\eta = -\frac{1}{g} \frac{\partial \phi}{\partial t}, \text{ for } y = H \quad (5.4)$$

For clearer visualization, the height of the wave formed at two fixed points on the front and back of the boat are plotted below

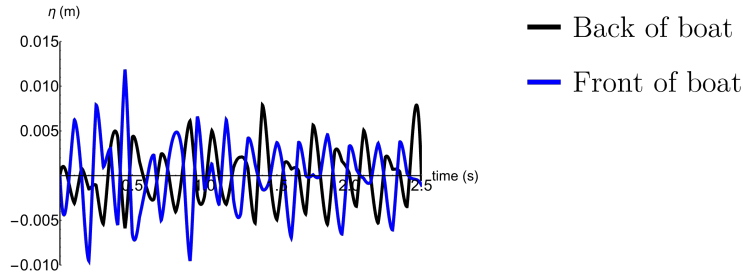


Figure 20: Height of the wave at 2 fixed points

A sample set of conditions is used to solve for the boat motion with and without accounting for the formation of gravity waves. When waves are accounted for, the depth of the boat in water is integrated across the surface. No significant deviation is observed, possibly because there is no net change in depth across the entire surface of the boat.

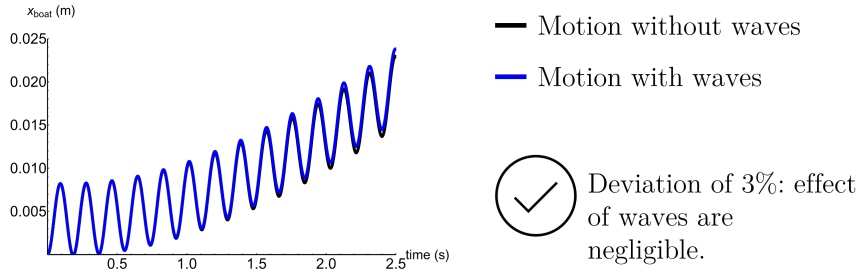


Figure 21: Boat trajectory with and without accounting for waves

6 Conclusion

Currently, an unequal angular frequency was achieved by exploiting gravity, which prevents the active varying of parameters. Hence, a better setup could include an Arduino controller to achieve greater control over the angular frequency.

In conclusion, in this research, a stiff-hulled boat that moves only due to the periodical mechanical movements of its internal parts was constructed. The boat motion was predicted both numerically and analytically, and phase plots were analyzed for greater physical insight. Extensive experiments were conducted to verify the quantitative models. Both the transient motion and the effect of varying parameters on the average velocity achieved by the boat were investigated, and the optimization of the boat was discussed. To achieve a greater average velocity, there should be a high mass of load to mass of boat ratio, high ratio of backward to forward drag coefficient, high forward angular velocity of load, and a certain backward angular velocity of load.

7 References

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8 Appendix

Here, a complete working for the steps outlined in Section 2.2 is given. Recall a single period T spanning $-\frac{\pi}{\omega_2} \leq t \leq \frac{\pi}{\omega_1}$, and

$$\dot{x}_{\text{boat}} \approx \begin{cases} v_o + \gamma\omega_2 \sin \omega_2 t, & -\frac{\pi}{\omega_2} \leq t < 0 \\ v_o + \gamma\omega_1 \sin \omega_1 t, & 0 \leq t < \frac{\pi}{\omega_1} \end{cases} \quad (8.1)$$

where $\gamma = \frac{mR}{M+m}$. The first condition is satisfied as the load moves forward, while the second condition is satisfied as the load moves backwards.

First, consider the case where the boat moves a net distance forward: $v_o > 0$. Let t_1 and t_2 be the time where $v = 0$. Since $v_o > 0$, this occurs as the load moves *forwards* (and the boat move backwards). Therefore, t_1 and t_2 are the solutions to $v_o + \gamma\omega_2 \sin \omega_2 t = 0$, for $-\frac{\pi}{\omega_2} \leq t_1 < t_2 \leq 0$.

$$t_2 = -\frac{1}{\omega_2} \sin^{-1} \frac{v_o}{\gamma\omega_2}$$

$$t_1 = -\frac{1}{\omega_2} \left(\pi - \sin^{-1} \frac{v_o}{\gamma\omega_2} \right) = -\frac{\pi}{\omega_2} - t_2$$

Analyzing the motion of the boat and load at each time interval,
 for $-\frac{\pi}{\omega_2} < t < t_1$ the load is moving forwards, and the boat is moving forwards.
 for $t_1 < t < t_2$ the load is moving forwards, and the boat is moving backwards.
 for $t_2 < t < 0$ the load is moving forwards, and the boat is moving forwards.
 for $0 < t < \frac{\pi}{\omega_1}$ the load is moving backwards, and the boat is moving forwards.

Therefore, the total drag force over a period is,

$$\begin{aligned} \int_{-\frac{\pi}{\omega_2}}^{\frac{\pi}{\omega_1}} F_D dt \approx & - \int_{-\frac{\pi}{\omega_2}}^{t_1} k_{\rightarrow} (v_o + \gamma\omega_2 \sin \omega_2 t)^2 dt + \int_{t_1}^{t_2} k_{\leftarrow} (v_o + \gamma\omega_2 \sin \omega_2 t)^2 dt \\ & - \int_{t_2}^0 k_{\rightarrow} (v_o + \gamma\omega_2 \sin \omega_2 t)^2 dt - \int_0^{\frac{\pi}{\omega_1}} k_{\rightarrow} (v_o + \gamma\omega_1 \sin \omega_1 t)^1 dt \end{aligned} \quad (8.2)$$

Knowing that

$$\int (v_o + \gamma\omega \sin \omega t)^2 dt = \left(v_o^2 + \frac{\gamma^2\omega^2}{2}\right)t - 2\gamma v_o \cos \omega t - \frac{\gamma^2\omega}{4} \sin 2\omega t$$

then,

$$\begin{aligned} \int_{-\frac{\pi}{\omega_2}}^{\frac{\pi}{\omega_1}} F_D dt &\approx -k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)t - 2\gamma v_o \cos \omega_2 t - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{-\frac{\pi}{\omega_2}}^{t_1} \\ &\quad + k_{\leftarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)t - 2\gamma v_o \cos \omega_2 t - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{t_1}^{t_2} \\ &\quad - k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)t - 2\gamma v_o \cos \omega_2 t - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{t_2}^0 \\ &\quad - k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_1^2}{2}\right)t - 2\gamma v_o \cos \omega_1 t - \frac{\gamma^2\omega_1}{4} \sin 2\omega_1 t \right]_{-0}^{\frac{\pi}{\omega_1}} \\ &= k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)(-t_2) - 2\gamma v_o \cos() - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{-\frac{\pi}{\omega_2}}^{t_1} \\ &\quad + k_{\leftarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)t - 2\gamma v_o \cos \omega_2 t - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{t_1}^{t_2} \\ &\quad - k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right)t - 2\gamma v_o \cos \omega_2 t - \frac{\gamma^2\omega_2}{4} \sin 2\omega_2 t \right]_{t_2}^0 \\ &\quad - k_{\rightarrow} \left[\left(v_o^2 + \frac{\gamma^2\omega_1^2}{2}\right)t - 2\gamma v_o \cos \omega_1 t - \frac{\gamma^2\omega_1}{4} \sin 2\omega_1 t \right]_{-0}^{\frac{\pi}{\omega_1}} \\ &= (k_{\leftarrow} + k_{\rightarrow}) \left(2 \left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right) t_2 - 4\gamma v_o \cos \omega_2 t_2 - \frac{\gamma^2\omega_2}{2} \sin 2\omega_2 t_2 \right) \\ &\quad + \left(v_o^2 + \frac{\gamma^2\omega_2^2}{2}\right) \frac{\pi k_{\leftarrow}}{\omega_2} - \left(v_o^2 + \frac{\gamma^2\omega_1^2}{2}\right) \frac{\pi k_{\rightarrow}}{\omega_1} \end{aligned} \tag{8.3}$$

Since v_o is small,

$$t_2 = -\frac{1}{\omega_2} \sin^{-1} \frac{v_o}{\gamma\omega_2} \approx -\frac{v_o}{\gamma\omega_2^2}$$

$$\cos \omega_2 t_2 = \cos \left(-\sin^{-1} \frac{v_o}{\gamma\omega_2} \right) \approx 1 - \frac{1}{2} \left(\frac{v_o}{\gamma\omega_2} \right)^2$$

$$\sin 2\omega_2 t_2 = \sin \left(-2 \sin^{-1} \frac{v_o}{\gamma\omega_2} \right) \approx -\frac{2v_o}{\gamma\omega_2}$$

Therefore,

$$\begin{aligned}
\int_{-\frac{\pi}{\omega_2}}^{\frac{\pi}{\omega_1}} F_D dt &\approx -(k_{\leftarrow} + k_{\rightarrow}) \left(2 \left(v_o^2 + \frac{\gamma^2 \omega_2^2}{2} \right) \frac{v_o}{\gamma \omega_2^2} \right. \\
&\quad \left. + 4\gamma v_o \left(1 - \frac{1}{2} \left(\frac{v_o}{\gamma \omega_2} \right)^2 \right) - \frac{\gamma^2 \omega_2}{2} \frac{2v_o}{\gamma \omega_2} \right) \\
&\quad + \left(v_o^2 + \frac{\gamma^2 \omega_2^2}{2} \right) \frac{\pi k_{\leftarrow}}{\omega_2} - \left(v_o^2 + \frac{\gamma^2 \omega_1^2}{2} \right) \frac{\pi k_{\rightarrow}}{\omega_1} \\
&= -k_{\leftarrow} \left(4\gamma v_o - \frac{\pi v_o^2}{\omega_2} - \frac{1}{2} \pi \gamma^2 \omega_2 \right) + k_{\rightarrow} \left(4\gamma v_o + \frac{\pi v_o^2}{\omega_1} + \frac{1}{2} \pi \gamma^2 \omega_1 \right) \\
&= -\pi \left(\frac{k_{\rightarrow}}{\omega_1} - \frac{k_{\leftarrow}}{\omega_2} \right) v_o^2 - 4\gamma (k_{\leftarrow} + k_{\rightarrow}) v_o - \frac{\pi \gamma^2}{2} (k_{\rightarrow} \omega_1 - k_{\leftarrow} \omega_2)
\end{aligned} \tag{8.4}$$

Since this drag averages to zero, the quadratic equation is solved for v_o ,

$$\begin{aligned}
v_o &= \frac{2\gamma \omega_1 \omega_2}{\pi} \frac{k_{\leftarrow} + k_{\rightarrow}}{k_{\leftarrow} \omega_1 - k_{\rightarrow} \omega_2} \left(1 \pm \sqrt{1 - \frac{\pi^2}{8\omega_1 \omega_2} \frac{(k_{\leftarrow} \omega_2 - k_{\rightarrow} \omega_1)(k_{\leftarrow} \omega_1 - k_{\rightarrow} \omega_2)}{(k_{\leftarrow} + k_{\rightarrow})^2}} \right) \\
&= \frac{2\gamma \omega_1 \omega_2}{\pi} \frac{k_{\leftarrow} + k_{\rightarrow}}{k_{\leftarrow} \omega_1 - k_{\rightarrow} \omega_2} \left(1 - \sqrt{1 - \frac{\pi^2}{8\omega_1 \omega_2} \frac{(k_{\leftarrow} \omega_2 - k_{\rightarrow} \omega_1)(k_{\leftarrow} \omega_1 - k_{\rightarrow} \omega_2)}{(k_{\leftarrow} + k_{\rightarrow})^2}} \right)
\end{aligned} \tag{8.5}$$

Since $v_o = 0$ when $k_{\leftarrow} = k_{\rightarrow}$ and $\omega_1 = \omega_2$, only the solution with the negative sign exists.